Forecasting and Macroeconomic Analytics

ECON 522

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Dynamic Causal Effects - (see SW)

Dynamic Causal Effects

A dynamic causal effect is the effect on Y of a change in X over time.

For example:

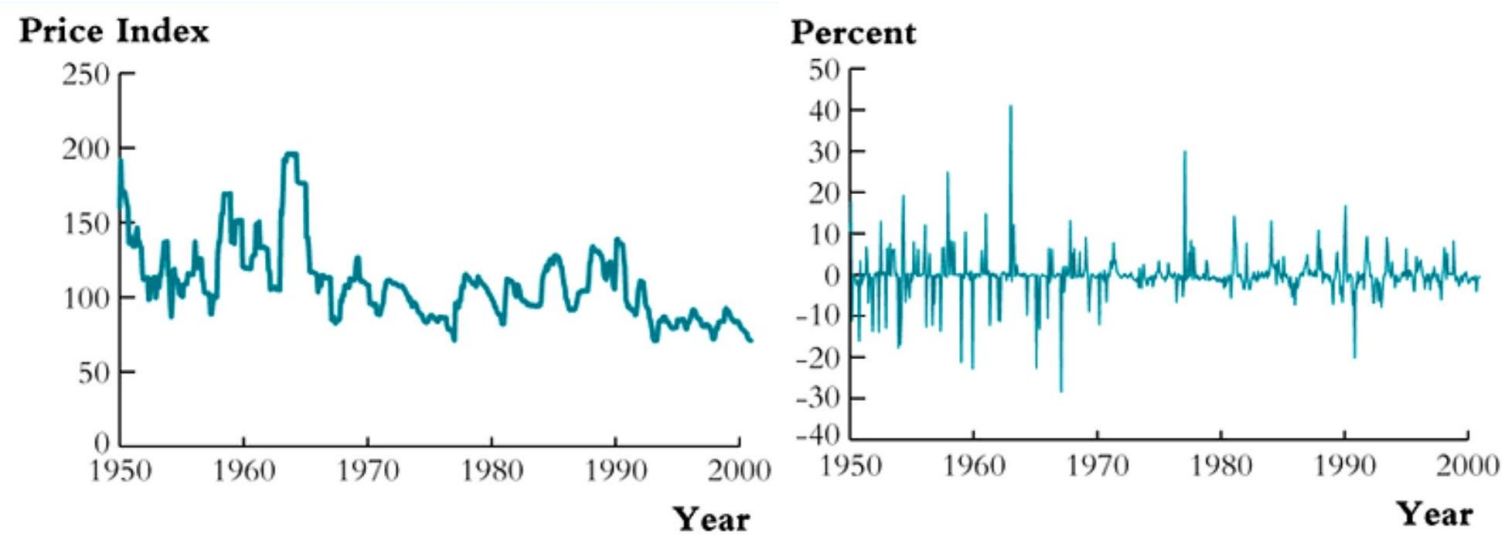
- The effect of an increase in cigarette taxes on cigarette consumption this year, next year, in 5 years;
- The effect of a change in the Fed Funds rate on inflation, this month, in 6 months, and 1 year;
- The effect of a freeze in Florida on the price of orange juice concentrate in 1 month, 2 months, 3 months...

The Orange Juice Data

- Monthly, Jan. 1950 - Dec. 2000 ($T = 612$). A little old data...
- Price = price of frozen OJ (a sub-component of the producer price index; US Bureau of Labor Statistics)
- $\%ChgP$ = percentage change in price overe a month, so $\%ChgP_t = 100\Delta \ln(\text{Price}_t)$
- FDD = number of freezing degree-days during the month, recorded in Orlando FL. Example: If November has 2 days with low temp below 32°, one at 30° and at 25°, then $FDD_{\text{Nov}} = 2 + 7 = 9$

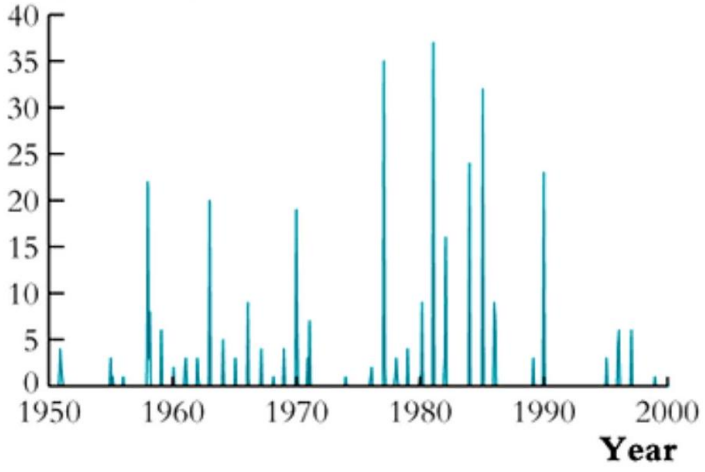
Data

Price and percent change i the price of frozen concentrate juice



There have been large month-to-month changes in the price of frozen concentrated orange juice.

Weather Data



Many of the large movements coincide with freezing weather in Orlando, home of the orange groves.

Initial Regression

$$\widetilde{\%Chg}_t = -.40 + .47FDD_t$$
$$(.22) \quad (.13)$$

- Statistically significant positive relation.: One additional freezing degree day increase prices by 0.47%
- More/deeper freezes, price goes up
- Standard errors: not the usual - heteroskedasticity and autocorrelation-consistent (HAC) SE's - more on this later
- But what is the effect of *FDD* over time?

Add dynamic to the OJ regression

$$\widehat{\%ChgP}_t = -\underset{(0.23)}{0.65} + \underset{(0.14)}{0.47}FDD_t + \underset{(0.08)}{0.14}FDD_{t-1} + \underset{(0.06)}{0.06}FDD_{t-2}$$
$$+\underset{(0.05)}{0.07}FDD_{t-3} + \underset{(0.03)}{0.03}FDD_{t-4} + \underset{(0.03)}{0.05}FDD_{t-5} + \underset{(0.04)}{0.05}FDD_{t-6}$$

- This is a distributed lags regression
- The estimates coefficients are the effect of a unit increase in current and future value of price changes: They are estimated of the dynamic effect of FDD.

Dynamic Causal Effects - Ideal

Example: What is the effect of fertilizer on tomato yield?

An ideal randomized controlled experiment

- Fertilize some plots, not others (random assignment)
- Measure yield over time - over repeated harvests - to estimate causal effect of fertilizer on:
- Yield in year 1 of expt
  - Yield in year 2 , etc.
- The result (in a large expt) is the causal effect of fertilizer on yield k years later.

Dynamic Causal Effects for OJ

In time series applications, we can't conduct this ideal randomized controlled experiment:

- We only have one US OJ market ....
- We can't randomly assign FDD to different replicates of the US OJ market (?)
- We can't measure the average (across "subjects") outcome at different times - only one "subject"
- So we can't estimate the causal effect at different times using the differences estimator

Dynamic Causal Effects for OJ

An alternative thought experiment:

- Randomly give the same subject different treatments (FDD_t) at different times
- Measure the outcome variable ($\%ChgP_t$)
- The "population" of subjects consists of the same subject (OJ market) but at different dates
- If the "different subjects" are drawn from the same distribution - that is, if Y_t, X_t are stationary - then the dynamic causal effect can be deduced by OLS regression of Y_t on lagged values of X_t.
- This estimator (regression of Y_t on X_t and lags of X_t) s called the distributed lag estimator.

Dynamic causal effects (DCE) and the distributed lag model

The distributed lag model is:

$$Y_t = \beta_0 + \beta_1 X_t + \ldots + \beta_p X_{t-r} + u_t$$

- β_1 = impact effect of change in $\boldsymbol{X}$ = effect of change in X_t on Y_t, holding past X_t constant
- β_2 = 1-period dynamic multiplier = effect of change in X_{t-1} on Y_t, holding constant $X_t, X_{t-2}, X_{t-3}, \ldots$
- β_3 = 2-period dynamic multiplier (etc.) = effect of change in X_{t-2} on Y_t, holding constant $X_t, X_{t-1}, X_{t-3}, \ldots$
- Cumulative dynamic multipliers
  oEx: the 2-period cumulative dynamic multiplier

$$= \beta_1 + \beta_2 + \beta_3 \text{ (etc.)}$$

Exogeneity in time series regression

Exogeneity (past and present) X is exogenous if
$E\left(u_t X_t, X_{t-1}, X_{t-2}, \ldots\right) = 0$.

Strict Exogeneity (past, present, and future) X is strictly exogenous if
$E\left(u_t \mid \ldots, X_{t+1}, X_t, X_{t-1}, \ldots\right) = 0$

- Strict exogeneity implies exogeneity
- For now we suppose that X is exogenous - we'll return (briefly) to the case of strict exogeneity later.
- If X is exogenous then OLS estimates the dynamic causal effect on Y of a change in X. Specifically,...

Estimation of DCEs with Exogenous Regressors

$$Y_t = \beta_0 + \beta_1 X_t + \ldots + \beta_{r+1} X_{t-r} + u_t$$

The Distributed Lag Model Assumptions

1. $E\left(u_t \mid X_t, X_{t-1}, X_{t-2}, \ldots\right) = 0$ (X is exogenous)
2. (a) Y and X have stationary distributions;
   (b) (Y_t, X_t) and (Y_{t-j}, X_{t-j}) become independent as j gets large
3. Y and X have eight nonzero finite moments
4. There is no perfect multicollinearity.

- Assumptions 1 and 4 are familiar
- Assumption 3 is familiar, except for 8 (not four) finite moments - this has to do with HAC estimators
- Assumption 2 is different - before it was (X_i, Y_i) are i.i.d. - this now becomes more complicated.

Estimation of DCEs with Exogenous Regressors

2. (a) Y and X have stationary distributions;

- If so, the coefficients don't change within the sample (internal validity);
- and the results can be extrapolated outside the sample (external validity).
- This is the time series counterpart of the "identically distributed" part of i.i.d.

2. (b) (Y_t, X_t) and (Y_{t-j}, X_{t-j}) become independent as j gets large

- Intuitively, this says that we have separate experiments for time periods that are widely separated.
- In cross-sectional data, we assumed that Y and X were i.i.d., a consequence of simple random sampling - this led to the CLT.
- A version of the CLT holds for time series variables that become independent as their temporal separation increases - assumption 2(b) is the time series counterpart of the "independently distributed" part of i.i.d.

Under the Distributed Lag Model Assumptions:

- OLS yields consistent estimators of $\beta_1, \beta_2, \ldots, \beta_r$ (of the dynamic multipliers)
- The sampling distribution of $\hat{\beta}_1$, etc., is normal
- However, the formula for the variance of this sampling distribution is not the usual one from crosssectional (i.i.d.) data, because u_t is not i.i.d. - it is serially correlated.
- This means that the usual OLS standard errors are wrong
- We need to use, instead, SE s that are robust to autocorrelation as well as to heteroskedasticity...

Heteroskedasticity and Autocorrelation-Consistent (HAC) Standard Errors

- When u_t is serially correlated, the variance of the sampling distribution of the OLS estimator is different.
- Consequently, we need to use a different formula for the standard errors.
- This is easy to do using most statistical software.

HAC the math

Consider the case of no lags:

$$Y_t = \beta_0 + \beta_1 X_t + u_t$$

"Recall" that the OLS estimator is:

$$\hat{\beta}_1 = \frac{\frac{1}{T}\sum_{t=1}^{T}\left(X_t - \bar{X}\right)\left(Y_t - \bar{Y}\right)}{\frac{1}{T}\sum_{t=1}^{T}\left(X_t - \bar{X}\right)^2}$$

$$\text{var}\left(\hat{\beta}_1\right) = \text{var}\left(\frac{1}{T}\sum_{t=1}^{T} v_t\right) / \left(\sigma_X^2\right)^2 \quad \text{(still SW App. 4.3)}$$

Variance for Time Series Data

What happens with time series data? Consider $T = 2$:

$$\begin{aligned}
\operatorname{var}\left(\frac{1}{T}\sum_{t=1}^{T} v_t\right) &= \operatorname{var}\left[1/2\left(v_1 + v_2\right)\right] \\
&= 1/4\left[\operatorname{var}\left(v_1\right) + \operatorname{var}\left(v_2\right) + 2\operatorname{cov}\left(v_1, v_2\right)\right] \\
\operatorname{var}\left(\frac{1}{2}\sum_{t=1}^{2} v_t\right) &= 1/4\left[\operatorname{var}\left(v_1\right) + \operatorname{var}\left(v_2\right) + 2\operatorname{cov}\left(v_1, v_2\right)\right] \\
&= 1/2\sigma_v^2 + 1/2\rho_1\sigma_v^2 \quad \left(\rho_1 = \operatorname{corr}\left(v_1, v_2\right)\right) \\
&= 1/2\sigma_v^2 \times f_2, \text{ where } f_2 = \left(1 + \rho_1\right)
\end{aligned}$$

- In i.i.d. (cross-section) data, $\rho_1 = 0$ so $f_2 = 1$ - which gives the usual formula for $\operatorname{var}\left(\hat{\beta}_1\right)$.
- In time series data, $\rho_1 \neq 0$, so $\operatorname{var}\left(\hat{\beta}_1\right)$ is not given by the usual formula
- Conventional OLS SE's are wrong when u_t is serially correlated.

Expression for $\text{var}\left(\hat{\beta}_1\right)$, general T

$$\text{var}\left(\frac{1}{T}\sum_{t=1}^{T} v\right) = \frac{\sigma_v^2}{T} \times f_T$$

so

$$\text{var}\left(\hat{\beta}_1\right) = \left[\frac{1}{T}\frac{\sigma_v^2}{\left(\sigma_X^2\right)^2}\right] \times f_T$$

where

$$f_T = 1 + 2\sum_{j=1}^{T-1}\left(\frac{T-j}{T}\right)\rho_j$$

The OLS SEs are off by the factor f_T (which can be big!)

HAC Standard Errors

- Conventional OLS SEs (heteroskedasticity-robust or not) are wrong when there is autocorrelation
- So, we need a new formula that produces SE s that are robust to autocorrelation as well as heteroskedasticity We need Heteroskedasticity and AutocorrelationConsistent (HAC) standard errors
- If we knew the factor f_T, we could just make the adjustment.
- But we don't know f_T - it depends on unknown autocorrelations.
- HAC SEs replace f_T with an estimator of f_T

HAC Standard Errors cn't

$$\text{var}\left(\hat{\beta}_1\right) = \left[\frac{1}{T}\frac{\sigma_v^2}{\left(\sigma_X^2\right)^2}\right] \times f_T, \text{ where } f_T = 1 + 2\sum_{j=1}^{T-1}\left(\frac{T-j}{T}\right)\rho_j$$

The most commonly used estimator of f_T is:

$$\hat{f}_T = 1 + 2\sum_{j=1}^{m-1}\left(\frac{m-j}{m}\right)\tilde{\rho}_j$$

- $\hat{f}_t$ sometimes called "Newey-West" weights
- $\tilde{\rho}_j$ is an estimator of ρ_j
- m is called the truncation parameter
- What truncation parameter to use in practice? -Use the Goldilocks method or try $m = 0.75\,T^{1/3}$

Example: OJ and HAC estimators

This is STATA output but it should not matter, L1* is the lag or order 1 of that variable.

```
Regression with robust standard errors                 Number of obs =     612
                                                       F(  1,   610) =    3.97
                                                       Prob > F      =  0.0467
                                                       R-squared     =  0.0101
                                                       Root MSE      =  5.0438

------------------------------------------------------------------------------
             |               Robust
        dlpoj |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
--------------+---------------------------------------------------------------
       l1fdd |   .1529217   .0767206     1.99   0.047     .0022532    .3035903
       _cons |  -.2097734   .2071122    -1.01   0.312    -.6165128     .196966
```

Example: OJ and HAC estimators

Now compute Newey-West SEs:

```
. newey dlpoj l1fdd if tin(1950m1,2000m12), lag(8);

Regression with Newey-West standard errors          Number of obs   =       612
maximum lag : 8                                     F(  1,    610)  =      3.83
                                                    Prob > F        =    0.0507

------------------------------------------------------------------------------
             |             Newey-West
       dlpoj |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
       l1fdd |   .1529217   .0781195     1.96   0.051     -.000494    .3063375
       _cons |  -.2097734   .2402217    -0.87   0.383    -.6815353    .2619885
```

Uses autocorrelations up to $m = 8$ to compute the SEs rule-of-thumb: $0.75^* \left(612^{1/3}\right) = 6.4 \approx 8$, rounded up a little.

OK, in this case the difference is small, but not always so!

HAC SE and AR

Do I need to use HAC SEs when I estimate an AR or an ADL model? Probably no but be careful.

- The problem to which HAC SEs are the solution arises when u_t is serially correlated
- If u_t is serially uncorrelated, then OLS SE's are fine
- In AR and ADL models, the errors are serially uncorrelated if you have included enough lags of Y
- If you include enough lags of Y, then the error term can't be predicted using past Y, or equivalently by past u - so u is serially uncorrelated

Estimation of Dynamic Causal Effects with Strictly Exogenous Regressors

- X is strictly exogenous if $E\left(u_t \mid \ldots, X_{t+1}, X_t, X_{t-1}, \ldots\right) = 0$
- If X is strictly exogenous, there are more efficient ways to estimate dynamic causal effects than by a distributed lag regression.
  -Generalized Least Squares (GLS)
- Autoregressive Distributed Lag (ADL)
- But the condition of strict exogeneity is very strong, so this condition is rarely plausible in practice.
- So we won't cover GLS or ADL estimation of dynamic causal effects (Section 13.5 is optional)

Analysis of the OJ Price Data

What is the dynamic causal effect (what are the dynamic multipliers) of a unit increase in FDD on OJ prices?

$$\%ChgP_t = \beta_0 + \beta_1 FDD_t + \ldots + \beta_{r+1} FDD_{t-r} + u_t$$

- What r to use?

How about 18? (Goldilocks method)

- What m (Newey-West truncation parameter) to use?

$$m = .75 \times 612^{1/3} = 6.4 \cong 7$$

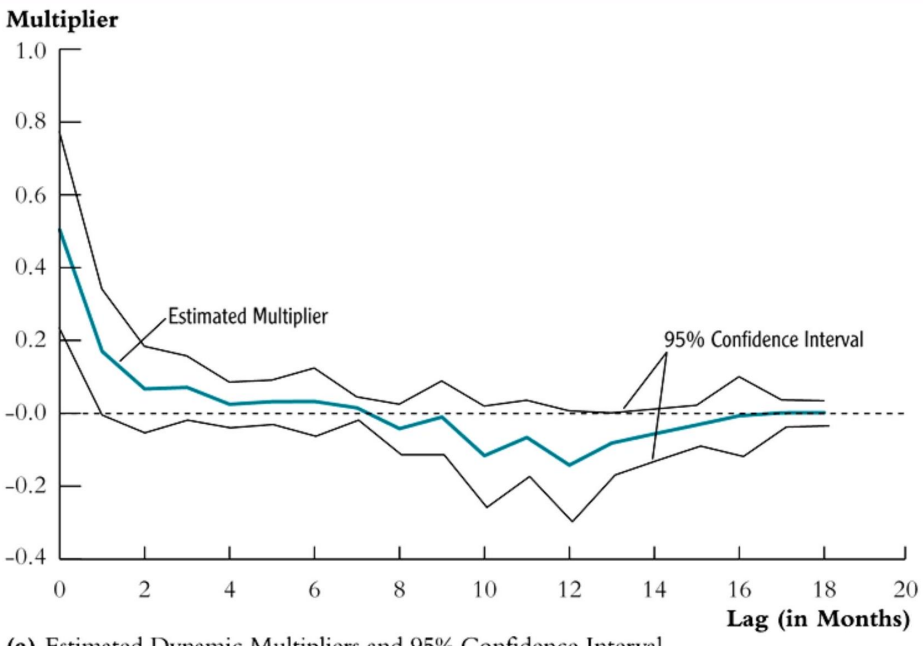
Estimated Multipliers

TABLE 13.1 The Dynamic Effect of a Freezing Degree Day (*FDD*) on the Price of Orange Juice: Selected Estimated Dynamic Multipliers and Cumulative Dynamic Multipliers

Lag number	(1) Dynamic Multipliers	(2) Cumulative Multipliers	(3) Cumulative Multipliers	(4) Cumulative Multipliers
0	0.50 (0.14)	0.50 (0.14)	0.50 (0.14)	0.51 (0.15)
1	0.17 (0.09)	0.67 (0.14)	0.67 (0.13)	0.70 (0.15)
2	0.07 (0.06)	0.74 (0.17)	0.74 (0.16)	0.76 (0.18)
3	0.07 (0.04)	0.81 (0.18)	0.81 (0.18)	0.84 (0.19)
4	0.02 (0.03)	0.84 (0.19)	0.84 (0.19)	0.87 (0.20)
5	0.03 (0.03)	0.87 (0.19)	0.87 (0.19)	0.89 (0.20)
6 ⋮	0.03 (0.05)	0.90 (0.20)	0.90 (0.21)	0.91 (0.21)
12 ⋮	−0.14 (0.08)	0.54 (0.27)	0.54 (0.28)	0.54 (0.28)
18	0.00 (0.02)	0.37 (0.30)	0.37 (0.31)	0.37 (0.30)
Monthly indicators?	No	No	No	Yes $F = 1.01$ $(p = 0.43)$
HAC standard error truncation parameter (m)	7	7	14	7

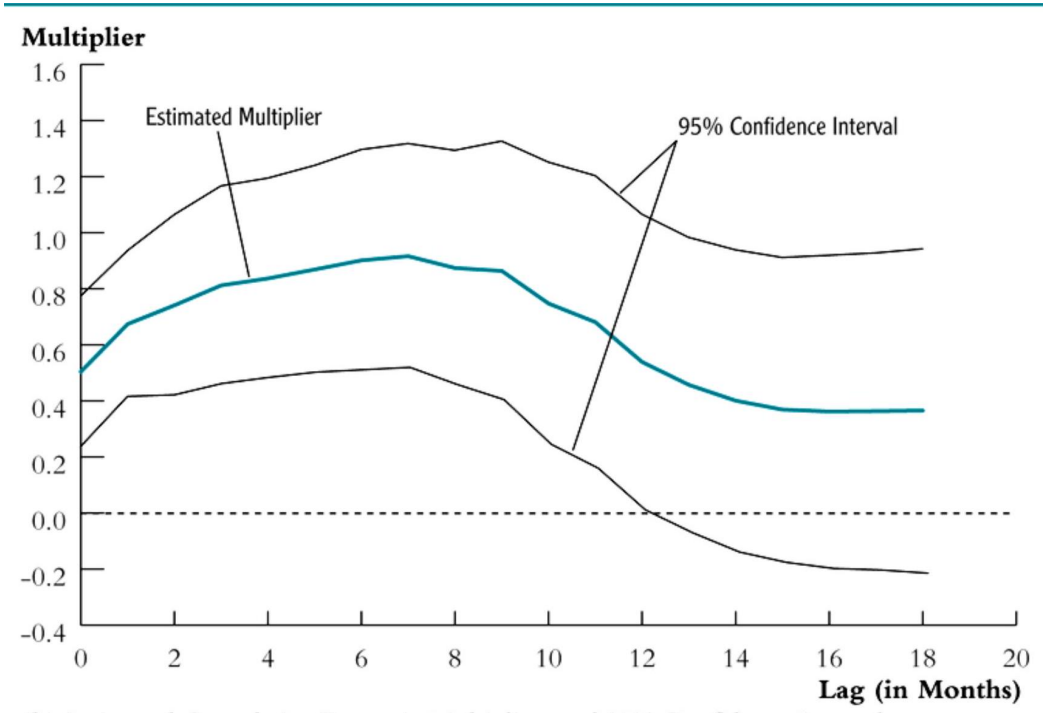
All regressions were estimated by OLS using monthly data (described in Appendix 13.1) from January 1950 to December 2000, for a total of $T = 612$ monthly observations. The dependent variable is the monthly percentage change in the price of orange juice ($\%ChgP_t$). Regression (1) is the distributed lag regression with the monthly number of freezing degree days and eighteen of its lagged values, that is, FDD_t, FDD_{t-1}, . . . , FDD_{t-18}, and the reported coefficients are the OLS estimates of the dynamic multipliers. The cumulative multipliers are the cumulative sum of estimated dynamic multipliers. All regressions include an intercept, which is not reported. Newey-West HAC standard errors, computed using the truncation number given in the final row, are reported in parentheses.

Mutiplier Graphic



The estimated dynamic multipliers show that a freeze leads to an immediate increase in prices. Future price rises are much smaller than the initial impact.

Mutiplier Graphic

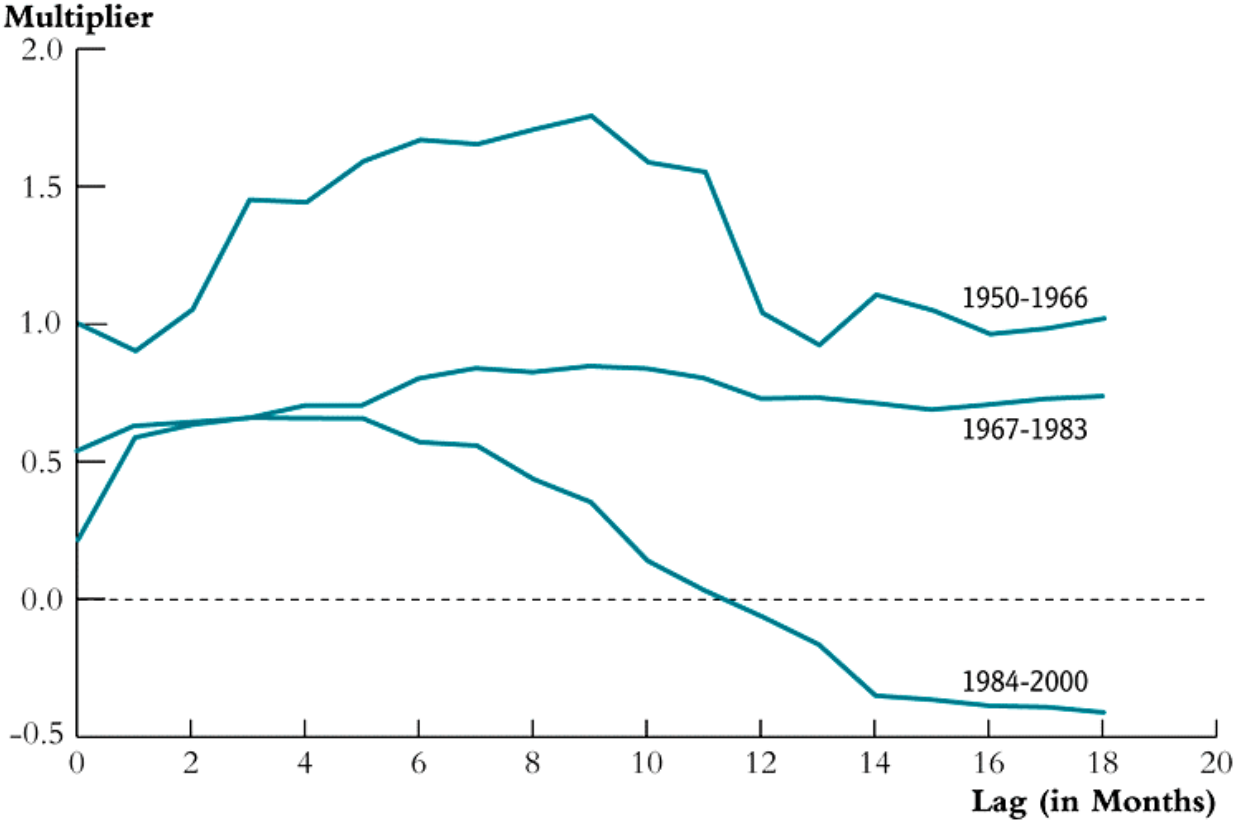


The cumulative multiplier shows that freezes have a persistent effect on the level of orange juice prices, with prices peaking seven months after the freeze.

Mutiplier Graphic- Different Periods

FIGURE 13.3 Estimated Cumulative Dynamic Multipliers from Different Sample Periods

The dynamic effect on orange juice prices of freezes changed significantly over the second half of the twentieth century. A freeze had a larger impact on prices during 1950 to 1966 than later, and the effect of a freeze was less persistent during 1984–2000 than earlier.



is FDD exogenous?

- Is *FDD* strictly exogenous in the distributed lag regression?

$$\%\,\text{Chg}_t = \beta_0 + \beta_1 FDD_t + \ldots + \beta_{r+1} FDD_{t-r} + u_t$$

- OJ commodity traders can't change the weather.
- So this implies that $\text{corr}\left(u_t, FDD_{t+1}\right) = 0$, right?

Thinking about exogeneity

That is, When is Exogeneity Plausible? In the following examples,

- is X exogenous?
- is X strictly exogenous?

Example:

1. Y = OJ prices, $X = FDD$ in Orlando
2. Y = Australian exports, X = US GDP (effect of US income on demand for Australian exports)

Thinking about exogeneity

Examples:

3. $Y = \mathrm{EU}$ exports, $X =$ US GDP (effect of US income on demand for EU exports)
4. $Y =$ US rate of inflation, $X =$ percentage change in world oil prices (as set by OPEC) (effect of OPEC oil price increase on inflation)
5. $Y =$ GDP growth, $X =$ Federal Funds rate (the effect of monetary policy on output growth)
6. $Y =$ change in the rate of inflation, $X =$ unemployment rate on inflation (the Phillips curve)

Exogeneity in Practice

- You must evaluate exogeneity and strict exogeneity on a case by case basis
- Exogeneity is often not plausible in time series data because of simultaneous causality
- Strict exogeneity is rarely plausible in time series data because of feedback.
- We will see this problem again when we discuss VAR. Identification in VAR is all about assuring that the shock of interest is exogenous.

Summary

- Dynamic causal effects are measurable in theory using a randomized controlled experiment with repeated measurements over time.
- When X is exogenous, you can estimate dynamic causal effects using a distributed lag regression
- If u is serially correlated, conventional OLS SEs are incorrect; you must use HAC SEs
- To decide whether X is exogenous, think hard!